

1) Create 'Reporting Exporter' through GRAPHITE in the Reporting tab in TrueNAS Scale:

- a. In the TrueNAS GUI, left hand navigation pane, click 'Reporting' -> top right corner, click 'Exporter' -> click 'Add'
- b. Fill in the field similar to this:

The screenshot shows the 'Edit Reporting Exporter' configuration window. The fields are as follows:

- Name: netdata
- Type: GRAPHITE
- Enable: ☒
- Destination IP: 192.168.1.100
- Destination Port: 9109
- Prefix: truenas
- Namespace: (empty)
- Update Every: 10
- Buffer On Failures: 10
- Send Names Instead Of IDs: ☒
- Matching Charts: *

- c. Your destination IP needs to be your actual TrueNAS IP address;
xxx.xxx.xxx.xxx
- d. Keep the destination port as 9109, that is how I got it to work
- e. I believe Prefix needs to be 'truenas' and choose another name for your Namespace, keep note of them.

- 2) Download 'graphite_mapping.conf' and 'netdata.conf' file from [Supporterino / truenas-graphite-to-prometheus](#) (you can also follow the directions there as well).
 - a. netdata.conf' will be used in step 3) and 'graphite_mnapping.conf will be used in step 4).
- 3) Copy 'netdata/conf' to /etc/netdata/netdata.conf
 - a. Helpful commands in CLI to get it to your respective location and privilege change:
 - i. `sudo cp /path/from/download/netdata.conf /etc/netdata/netdata.conf`
 - ii. `sudo chown root:root /etc/netdata/netdata.conf`
 - iii. `sudo chmod 644 /etc/netdata/netdata.conf`
 - iv. Restart Netdata - `sudo systemctl restart netdata`
 - v. Verify Netdata is running - `systemctl status netdata`
- 4) Install 'graphite-exporter' on your TrueNAS Scale as a custom app in TrueNAS GUI:
 - a. Screenshots to help, in order of what you see when you are populating the information, then instructions below the images.

Applications

Edit Custom App

Application name

Application Name *

graphite-exporter

General

Notes ⓘ

1

2

3

Image Configuration

Image ⓘ

Repository * ⓘ

ghcr.io/supporterino/truenas-graphite-to-prometheus ⓘ

Tag ⓘ

latest ⓘ

Pull Policy * ⓘ

Pull the image if it is not already present on the host.

Network Configuration

☐ Host Network ⓘ

Ports ⓘ

Add

Port Bind Mode ⓘ

Publish port on the host for external access

Host Port *

9109 ⓘ

Container Port *

9109 ⓘ

Protocol *

TCP

Host IPs ⓘ

Add

Host IP *

Port Bind Mode ⓘ

Publish port on the host for external access

Host Port *

9108 ⓘ

Container Port *

9108 ⓘ

Protocol *

TCP

Host IPs ⓘ

Add

Host IP *

Storage Configuration

Storage ⓘ

Add

Type * ⓘ

Host Path (Path that already exists on the system)

☒ Read Only ⓘ

Mount Path * ⓘ

/etc/graphite_exporter/graphite_mapping.conf ⓘ

Host Path Configuration

☐ Enable ACL ⓘ

Host Path * ⓘ

/mnt

Create Dataset

- b. Left hand navigation pane, click 'Apps' -> top right corner, click 'Discover Apps' -> click 'Custom App'.
- c. Fill in the following mandatory fields:
 - i. Image Configuration:
 1. Image Repository: `ghcr.io/supporterino/truenas-graphite-to-prometheus`
 2. Tag: Latest
 3. Pull Policy: Pull the image if it is not already present on the host.
 - ii. Network Configuration:
 1. Port Blind Mode: Publish port on the host for external access
 2. Add x1 -> Host Port: 9109; Container Port: 9109; Protocol: TCP; Host IP: Your TrueNAS IP xxx.xxx.xxx
 3. Add x2 -> Host Port: 9108; Container Port: 9108; Protocol: TCP; Host IP: Your TrueNAS IP xxx.xxx.xxx
 - iii. Storage – I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
 1. Type: Host Path (Path that already exists on the system)
 2. Read Only: Check
 3. Mount Path:


```
/mnt/path/to/respective/directory/graphite_mapping.conf
```

 - a. As noted in step 2) copy over the 'graphite_mapping.conf' file download from Supporterino / truenas-graphite-to-prometheus and place it in that repository which you will point to here.
 - b. Helpful commands in CLI to get it to your respective location and privilege change:
 - i. `sudo cp`
`/path/from/download/location/graphite_mapping.conf`
`/path/to/repo/graphite_mapping.conf`
 - ii. `sudo chmod 644`
`/path/from/download/location/graphite_mapping.conf`
 - iv. Click the 'Update' button at the bottom.
- d. Test in Shell -> `curl -s http://youripaddress:9108/metrics | grep truenas_cpu`

- 5) Install 'prometheus' on your TrueNAS Scale through 'Discover Apps':
 - a. Screenshots to help, in order in what you see when you are populating the information on the next page, then instructions below the images

Application name

Application Name *

prometheus

Prometheus Configuration

Retention Time *

4w

Retention Size

40GB

☒ WAL Compression

Additional Arguments

Add

No items have been added yet.

Additional Environment Variables

Add

No items have been added yet.

Network Configuration

WebUI Port

Port Bind Mode

Publish port on the host for external access

Port Number *

30104

Host IPs

Add

No items have been added yet.

☒ Host Network

Storage Configuration

Prometheus Config Storage

Type *

Host Path (Path that already exists on the system)

Host Path Configuration

☐ Enable ACL

Host Path *

/mnt

Create Dataset

Prometheus Data Storage

Type *

Host Path (Path that already exists on the system)

Host Path Configuration

☐ Enable ACL

Host Path *

/mnt

Create Dataset

- b. Left hand navigation pane, click 'Apps' -> top right corner, click 'Discover Apps'
- c. Search 'Prometheus'
- d. Click 'Install' and fill the below
 - i. Application Name:
 1. Application Name: Prometheus
 - ii. Prometheus Configuration:
 1. Retention Time – pick to your liking, I have 4w (w-weeks, d-days, m-months, etc.)
 2. Retention Size – pick to your liking, I have 40GB
 - iii. Network Configuration:
 1. Publish port on the host for external access
 2. Port Number: 30104
 - iv. Storage Configuration I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
 1. Prometheus Config Storage:
 - a. Type: Host Path (Path that already exists on the system)
 - b. Mount Path: /mnt/path/to/respective/directory
 2. Prometheus Data Storage:
 - a. Type: Host Path (Path that already exists on the system)
 - b. Mount Path: /mnt/path/to/respective/directory
 - v. Click the 'Update' button at the bottom.
 - vi. I had to navigate to the prometheus.yaml in Shell and update the yaml file. It did not auto-populate for some reason:
 1. Navigate to shell:
 2. Navigate to where you had your 'Prometheus' app installed and you should see a prometheus.yaml file.
 - a. Edit it by `sudo nano prometheus.yaml`
 - b. Leverage the below code into the file and save

```
global:
  scrape_interval: 10s
scrape_configs:
  - job_name: "truenas_graphite_exporter"
    static_configs:
      - targets: ["xxx.xxx.xxx.xxx:9108"]
    labels:
      instance: "truenas"
```

```
env: "xxxxx"
```

The “instance” and “env” should tie to step 1) where you define the “prefix” and “namespace” (red text above)

- vii. Once the ‘Prometheus App’ status is running, click on it and Application Info side window should populate.
 - viii. Click on ‘Web UI’ -> a dialogue box should pop up, enable HTTPS Redirect, click ‘Continue’
 - 1. It should open a browser into Prometheus UI.
 - 2. On the top banner, click Status -> then Target Health
 - 3. In the Endpoint section, you should see your TrueNAS scale IP address
 - 4. Click it and you can see all the data scrapes in it in text form.
- 6) Lastly, install ‘grafana’ on your TrueNAS Scale through ‘Discover Apps’:
- a. Screenshots to help, in order in what you see when you are populating the information on the next page, then instructions below the images

Edit Grafana

Application name

Application Name *

grafana

Network Configuration

WebUI Port ?

Port Bind Mode ?

Publish port on the host for external access

Port Number *

30037

Host IPs ?

Add

No items have been added yet.

☒ Host Network ?

Certificate ID ?

Search

Storage Configuration

Grafana Data Storage ?

Type *

Host Path (Path that already exists on the system)

Host Path Configuration

☐ Enable ACL ?

Host Path *

/mnt

Create Dataset

Grafana Plugins Storage ?

Type *

Host Path (Path that already exists on the system)

Host Path Configuration

☐ Enable ACL ?

Host Path *

/mnt

Create Dataset

Additional Storage

Add

No items have been added yet.

- b. Left hand navigation pane, click 'Apps' -> top right corner, click 'Discover Apps'
- c. Search 'Grafana'
- d. Click 'Install' and fill the below:
 - i. Application Name:
 1. Application Name: Grafana
 - ii. Network Configuration:
 1. Web UI
 2. Publish port on the host for external access
 3. Port Number: 30037
 - iii. Storage Configuration I have a pool dedicated to Apps, so this is how I did it (your milage may vary):
 1. Grafana Data Storage
 - a. Type: Host Path (Path that already exists on the system)
 - b. Host Path: /mnt/path/to/respective/directory
 2. Grafana Plugins Storage:
 - a. Type: Host Path (Path that already exists on the system)
 - b. Host Path: /mnt/path/to/respective/directory
 - iv. Click the 'Update' button at the bottom.
 - v. Once the 'Grafana App' status is running, click on it and Application Info side window should populate.
 - vi. Click on 'Web UI' -> a dialogue box should pop up, enable HTTPS Redirect, click 'Continue'
 1. It should open a browser into Grafana UI.
 2. On the left banner, click Connections -> search Prometheus
 3. Click 'Add new data source' in the top right corner
 4. It should automatically find your Prometheus App
 5. You can then navigate to 'Dashboards' in the left banner and click 'New' in the top right corner -> 'New Dashboard'.
 6. It should then automatically suggest the Prometheus data source for you to select.

You should now be set to create dashboards and begin monitoring your TrueNAS SCALE system.

Here is a sample of what I have created, the below dashboard image is still work in process:

